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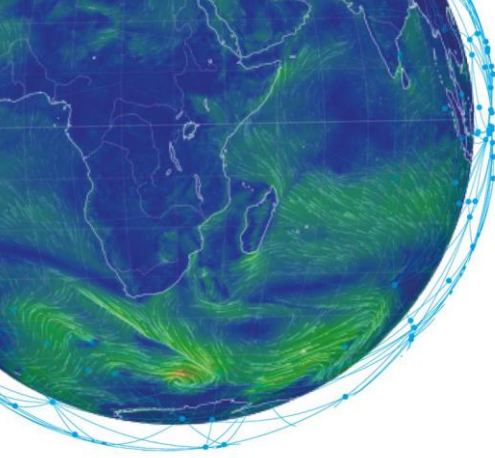
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Machine Learning in Weather & Climate

TRANSCRIPT

**State-of-the-art and challenges of
using observations in NWP**

Expert: Alan Geer



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Introduction

Hello and welcome to this 5-minute introduction video about the state of the art and challenges of using observations in NWP. My name is Lisa Burke and today, we'll meet Alan Geer, who is an expert on using satellite observations for weather forecasting. He works in the satellite microwave observations team at ECMWF, with a focus on using the data in areas of cloud and rain.

So, let's go to ECMWF headquarters in Reading in the UK.

Script

In weather and climate studies, and in machine learning, we make models to represent and predict the world around us. We find out about the world by making observations, but using those observations is surprisingly hard.

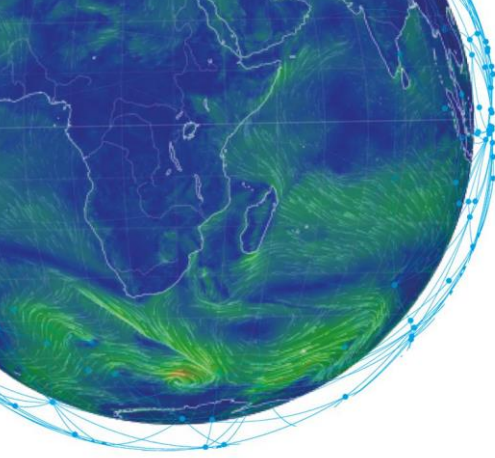
For example, how much rain is falling? We could put out a can and weight it after an hour – that's how typical rain gauges work.

But what if it's windy? Some of the rain might miss the can entirely.

And what if the sun comes out before the hour is up? Some of the water in the can will evaporate.

And in showery weather the rain might be falling somewhere else.

These are all examples of errors in our measurement process. Some of these errors are random, which might include the precision of the weighing scale, and some are systematic, which would include things like wind-driven undercatch and evaporation. Understanding measurement errors is key to making use of observations in weather forecasting.



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Another way to find out the quantity of rain is using microwave sensors that are flown on a number of satellites. In this picture we see the measurements from the Advanced Microwave Scanning Radiometer-2 on JAXA's GCOM-W satellite. As the satellite has passed overhead, it has made observations over a swath about 1500 km wide. The islands in the northern part of the picture include Sumatra and Java. The instrument measures the naturally emitted radiation of the earth in the microwave part of the spectrum, at different frequencies - 19 GHz and 89 GHz here.

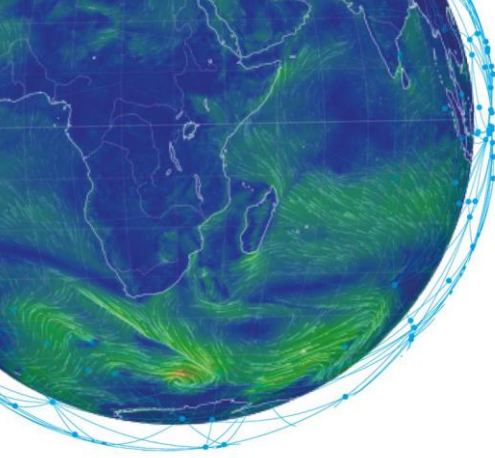
Over the ocean south of Sumatra, the grey spots at 19 GHz are relatively bright scenes caused by the emission of microwave radiation from raindrops. Over Sumatra itself, there are dark spots generated by the upper parts of deep convective clouds, and there is probably rain beneath.

To use a measurement like AMSR2 to improve a weather forecast model, we need to develop a physical model of the measurement process, in this case the natural processes that generate the radiation emitted by the earth, here including the effect of raindrops and snow particles. This is called an observation operator or sometimes an 'observation model'.

With the rain gauge example we also saw how a knowledge of the statistical properties of the errors in the measurement process is really important. These two things, along with the observations themselves, are what we use to update our models to better fit reality. We meet the observations half way. The observations give us a partial view of the system we want to model, and the model gives a partial reconstruction of the observations. The technical process for doing this is called data assimilation.

To finish off this quick survey, we can take a look at the observations that are used at ECMWF. On one side are the so called in-situ observations, like measurements of atmospheric temperature and pressure made from surface stations, including ships and buoys. This figure shows the relative number of these observations used at ECMWF in the atmospheric part of the weather prediction system during the month of August. Measurements are sparse or non-existent over ocean areas and some land areas too. This is one reason we need a diversity of observations, including those from satellites, which typically have a more global coverage.

Finally, this pie chart summarises the full range of observation types assimilated at ECMWF, broken down using an approximate measure of the benefit of each observation type to the quality of our forecasts. There is a wide range of mostly in-situ measurements from aircraft and other types of ground-based platforms and stations. In terms of satellite data, as well as the microwave radiances that we looked at just now, there are also infrared radiances, and techniques including GPS radio occultation, scatterometry, atmospheric motion vectors and wind lidars. We rely on all of these observations to be able to make our forecasts.



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Summary

We've been looking at how to improve our weather forecast model, which we want to bring closer to reality, through machine learning training, or through data assimilation. We ultimately have to do that by taking observations. Weather forecasting systems use observations from diverse sources including satellites, aircraft, balloons and ground stations. But because these measurements are typically indirect, sparse, and have errors, a range of special techniques are needed to use them.

Thank you Alan Geer, for this brief overview about the state of the art and challenges of using observations in NWP. If you have any question or would like to discuss this subject in further detail, please use our Forum on Moodle. It will be a pleasure to hear from you.